

Multiresponsive Shape-Adaptable Phase Change Materials with Cellulose Nanofiber/Graphene Nanoplatelet Hybrid-Coated Melamine Foam for Light/Electro-to-Thermal Energy Storage and Utilization

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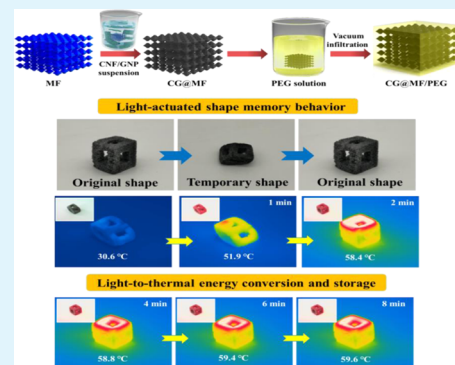
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Supporting Information

ABSTRACT: Strong rigidity, low thermal conductivity, and short of multi-driven capabilities of form-stable phase change materials (FSPCMs) have limited their practical utilization. Herein, we report a shape-adaptable FSPCM with the coinstantaneous light/electro-driven shape memory properties and light/electro-to-thermal energy storage performance. The FSPCM is fabricated by incorporating the poly(ethylene glycol) (PEG) into the cellulose nanofiber/graphene nanoplatelet (GNP) hybrid-coated melamine foam (CG@MF). The CG@MF/PEG FSPCMs show a good encapsulation effect, enhanced thermal conductivity, and large melting enthalpy (178.9 J g^{-1}). Due to the high elasticity of MF and the excellent photothermal conversion and electrical conductivity of the GNP network, the CG@MF/PEG FSPCMs exhibit a remarkable light/electro-driven shape memory effect by activating the phase change process of PEG. Meanwhile, the CG@MF/PEG FSPCMs can effectively convert light or electric energy into heat energy and reposit the converted energy during the phase change process. Furthermore, the CG@MF/PEG FSPCMs possess excellent multiresponsive self-adhesion properties. A light-sensitive, shape-adaptable, and thermal-insulating container is further explored. This study provides routes toward the development of multiresponsive shape-adaptable FSPCMs for energy storage applications.

KEYWORDS: multiresponsive, phase change, thermal conductivity, shape memory properties, shape-adaptable container



1. INTRODUCTION

Thermal energy storage is one of the important strategies for improving the energy storing efficiency and plays a critical role in the relevant thermal management applications such as the solar energy storage system, thermal protection of electronics, comfortable clothes, energy-saving buildings, etc.¹ Phase change materials (PCMs) are typical thermal energy storage materials that have received much attention due to their high latent heat and energy storage density.^{2,3} To prepare form-stable PCMs (FSPCMs) that can effectively prevent the liquid leakage during phase transition, various strategies, including porous absorption, microencapsulation, coaxial electrospinning, and polymer blending, are proposed.^{4,5} However, the FSPCMs always show strong rigidity, which inevitably leads to the installation difficulty, brittle fracture, and poor surface contact with devices.^{6,7} Therefore, the preparation of FSPCMs with shape variation ability has a significant meaning for practical applications. With this in mind, polymers with flexibility are chosen as supporting materials to prepare flexible FSPCMs. For example, some thermoplastic elastomers and polymer foams [e.g. melamine foam (MF)] are blended with

paraffin, and these FSPCMs possess good shape memory properties.^{8–11}

Although flexible FSPCMs are good materials for thermal energy storage, their advanced utilization has been limited by three challenges: (a) low thermal conductivity that restricts the heat transfer rate;^{12–14} (b) poor light-driven shape change and light-to-thermal energy conversion capabilities due to the weak absorption of light;^{15,16} (c) low electrical conductivity that makes FSPCMs hardly convert electric energy into heat energy.^{17–19} The introduction of carbon nanofillers like graphene and carbon nanotubes (CNTs) has been demonstrated to be a facile way to enhance the thermal conductivity, light-to-thermal, and electro-to-thermal convention performance.^{20–25} Among these carbon nanofillers, graphene nanoplatelets (GNPs), as one of the allotropes of carbon-based materials, have attracted great research interest and wide application in both academic and industrial fields.^{26,27} GNPs

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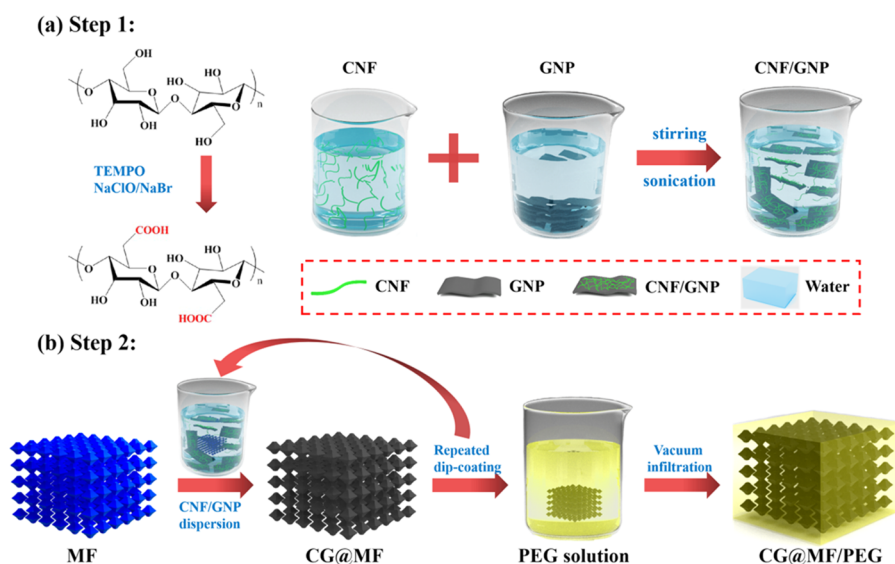


Figure 1. Schematic diagram showing the preparation process of the CNF/GNP hybrids and the CG@MF/PEG PCMs. (a) Preparation of the well-dispersed CNF/GNP hybrid suspension. (b) Preparation of the CG@MF sponges by repeated dip-coating and the CG@MF/PEG PCMs by vacuum infiltration method.

possess some intriguing properties, such as ultrahigh mechanical properties, excellent thermal conductivity, photo-thermal conversion, electrical conductivity, etc.²⁸ However, due to the chemical inertness, GNPs are difficult to disperse homogeneously in solvent and form a percolated conductive network in PCMs. Therefore, the large scale and environmentally friendly methods to achieve a three-dimensional (3D) interconnected GNP network are urgently required.

Recently, melamine foam (MF) has been intensively used as a building scaffold to load other functional materials.²⁹ Because of the low density, high porosity, and high elasticity, MF is often used to achieve 3D interconnected carbon-based materials, which can be applied in the oil/water separation, pressure sensor, electrochemistry, etc.^{30–35} In addition, cellulose nanofiber (CNF), which is derived from natural cellulose fibers through specific treatment, has become attractive because of its unique features: large aspect ratio, high mechanical strength, and good aqueous stability.^{36,37} Some studies have pointed that CNF is an ideal dispersant for the dispersion of carbon nanofillers including expanded graphite, CNTs, and graphene.^{38–41} More importantly, unlike small molecular surfactants or polymeric dispersants, CNF is capable of assisting the dispersion of carbon nanofillers without destroying the intrinsic properties of carbon nanofillers.⁴² Therefore, it is expected that MF works as a framework, and CNF works as a dispersant for GNP to prepare GNP@MF sponges with light/electro-responsibility.

Herein, we employ the CNF/GNP hybrid-coated MF sponge (CG@MF) as a flexible supporting scaffold to fabricate poly(ethylene glycol) (PEG)-based PCMs. CNF was first used to disperse GNPs, and then the CNF/GNP hybrid suspension was impregnated with MF via a dip-coating method. The PEG infiltrated CG@MF PCMs exhibit good form stability, large phase change enthalpy (178.9 J g⁻¹), and thermal-driven shape memory properties (the shape fixing ratio and the shape recovery ratio are all almost 100%). Furthermore, the GNP network assembled on MF imparts the CG@MF/PEG PCMs with enhanced thermal conductivity, light/electro-to-thermal energy storing capability, and light/electro-driven shape

memory properties. Interestingly, it is found that the CG@MF/PEG PCMs possess excellent self-adhesion properties under three stimuli (heat, light, and electricity). The multifunctional PCMs are considered to have a broad application prospect in lots of energy storage devices.

2. EXPERIMENTAL SECTION

2.1. Materials. GNPs (with a purity of 95%) were purchased from Deyang Carbonene Co., Ltd. (China). Microfibrillated cellulose (MFC, solid content, 25%) was purchased from Daicel Chemical Industries, Ltd. (Japan). MF with a density of 0.025 g cm⁻³ was purchased from Xiamen Share Nano Technology Co., Ltd. (China). PEG (6000 g mol⁻¹, *T_m* = 55–59 °C) was purchased from Kelong Chemical Reagent Co., Ltd. (China).

2.2. Material Preparation. **2.2.1. Preparation of CNF and CNF/GNP Hybrid Suspension.** CNF was produced using MFC as the raw material via a (2,2,6,6-tetramethylpiperidin-1-yl)oxyl (TEMPO)-oxidized method.³⁹ For the CNF/GNP hybrids, GNP powder (0.9 g) was added into 100 mL of CNF aqueous suspension (containing 0.1 g of CNF), and then the CNF/GNP hybrid suspension was treated by sonication for 1 h under the power of 300 W.

2.2.2. Preparation of CG@MF Sponges. A cubic MF sponge (size of 3 × 2 × 2 cm³) was immersed in the CNF/GNP hybrid suspension (2 mg mL⁻¹), and then the MF sponge was squeezed to make the CNF/GNP hybrids to wrap on the skeletons of MF. The CNF/GNP hybrid-coated MF (CG@MF) sponges were obtained after drying in an oven at 60 °C for 24 h. Furthermore, the content of CNF/GNP hybrids in the sponges could be adjusted through the repeated dip-coating method. Herein, three kinds of CG@MF sponges were prepared, namely, CG1@MF, CG3@MF, and CG5@MF. The numbers (1, 3, and 5) represent the dip-coating cycles.

2.2.3. Preparation of CG@MF/PEG PCMs. First, 40 g of PEG was added into deionized water (100 mL) at 65 °C for 1 h, and a completely molten PEG solution was obtained (0.4 g mL⁻¹). Then, under vacuum conditions, the CG@MF sponges were added into the molten PEG solution to mix for 10 min. Finally, the CG@MF/PEG PCMs were prepared by placing them in an oven at 50 °C to evaporate water. In addition, MF/PEG PCM was fabricated in the same way for performance comparison. Herein, the weight ratio of MF or CG@MF sponge to PEG was fixed at 1:20. In this work, the CNF/GNP hybrid contents in the CG@MF/PEG PCMs were 0.19 wt % (CG1@MF/PEG), 0.43 wt % (CG3@MF/PEG), and 0.65 wt % (CG5@MF/PEG), respectively. The preparation process of the

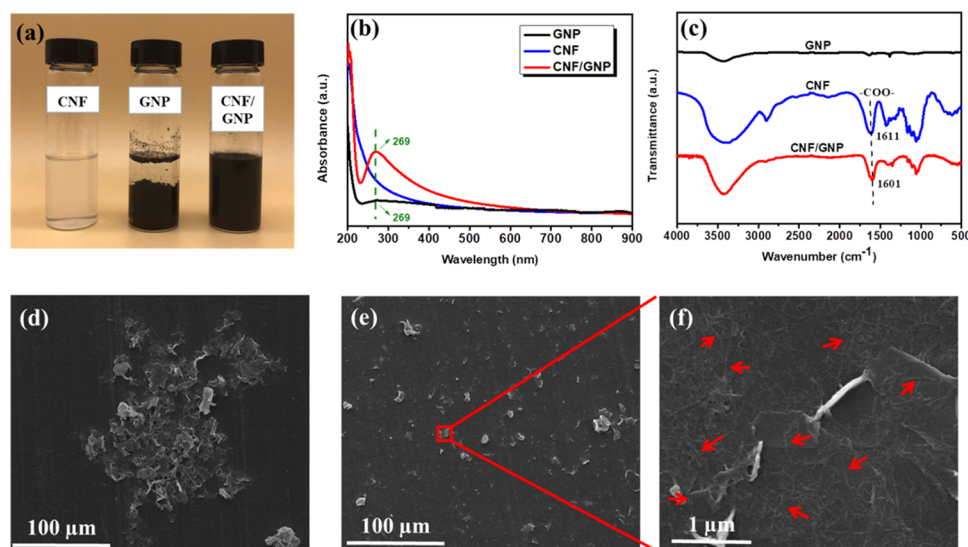


Figure 2. (a) Digital photographs of the CNF, GNP, and CNF/GNP suspensions in deionized water. (b) UV–vis absorption curves of the CNF, GNP, and CNF/GNP suspensions. (c) FTIR spectra of the CNF, GNP, and CNF/GNP powders. (d) SEM image of the neat GNP. (e, f) SEM images of the CNF/GNP hybrid.

CNF/GNP hybrids and the CG@MF/PEG PCMs is illustrated in Figure 1.

2.3. Structural Characterization and Performance Measurements. The structures of CNF, GNP, and CNF/GNP were investigated by the Fourier transform infrared (FTIR) spectroscopy (Thermo Nicolet) and UV–vis absorbance spectroscopy (UV-2600, Japan). The morphology of CNF was investigated by a transmission electron microscope (TEM, JEM-2100F, Japan) at an operating voltage of 200 kV and an atomic force microscope (AFM, HE002-H) in a tapping mode. Scanning electron microscope (SEM, Fei Inspect, Netherlands) was used to investigate the morphologies of GNP, CNF/GNP, MF, and CG@MF sponges. A DHR-1 rheometer (TA) was utilized to test the compression–recovery performances of CG@MF sponges.

The morphologies of MF/PEG and CG@MF/PEG PCMs were observed by SEM under an accelerating voltage of 20 kV. The structure of PEG, MF/PEG, and CG@MF/PEG PCMs were investigated by FTIR and a wide-angle X-ray diffractometer (WAXD, PANalytical Empyrean). The X-ray source was Ni-filtered Cu K α radiation, and the incident angle for the measurement was 5–50°. A DHR-1 rheometer (TA) was utilized to characterize the shape stabilities of MF/PEG and CG@MF/PEG PCMs. Under a heating rate of 5 °C min^{−1} and a compressed force of 0.3 N, the specimens were gradually heated from 30 to 100 °C. The dimension retention ratios of specimens were obtained according to eq 1

$$\text{dimension retention ratio} = \frac{L_2}{L_1} \times 100\% \quad (1)$$

where L_1 is the initial thickness of the specimen, and L_2 is the real-time thickness of the specimen during the heating process.

The differential scanning calorimeter (DSC, TA 2500) was used to evaluate the thermal–physical properties of MF/PEG and CG@MF/PEG PCMs. The specimens were first heated from 0 to 100 °C with a heating rate of 5 °C min^{−1} and immediately kept at 100 °C for 5 min, and then cooled to 0 °C with a cooling rate of 5 °C min^{−1}. The latent heat retention ratio was calculated according to eq 2

$$\text{latent heat retention ratio} = \frac{\Delta H_{c2}}{\Delta H_{c1}} \times 100\% \quad (2)$$

where ΔH_{c1} represents the crystallization enthalpy of pure PEG, and ΔH_{c2} represents the crystallization enthalpy of MF/PEG or CG@MF/PEG.

The thermal conductivities of MF/PEG and CG@MF/PEG PCMs were tested by a Transient Hot Disk TPS 2500S instrument (Sweden) according to ISO/CD 22007-2. The heat transfer behaviors of MF/PEG and CG@MF/PEG PCMs on a hotplate (80 °C) were recorded by a FLIR T620 infrared thermography camera.

A bend-recovery measurement was designed to evaluate the shape memory properties of MF/PEG and CG@MF/PEG PCMs. It was divided into four steps. (1) Heating the specimens to 60 °C and making the specimens change from a straight shape to a bent shape (180°). (2) Applying a constant force to fix the deformed samples and placing it in liquid nitrogen for a few seconds. (3) Releasing the external constant force and then recording the fixation of temporary shape at 25 °C. (4) Reheating the specimens to 60 °C and using a digital camera to record the shape recovery behaviors. For the shape memory process under thermal/light stimulation, a hotplate (80 °C) and a near-infrared (NIR) lamp (Philips, Netherlands) were utilized as heat and a light source, respectively. The distance between NIR lamp and the specimen was set at 35 cm, and the density of light irradiation was about 20 mW cm^{−2}. For electro-driven shape memory behavior, a direct current (DC) power supply (RXN-603D, China) was utilized as the power source, and the applied voltage was 40 V. By referring to the following eqs 3 and 4, the shape fixing ratio (R_f) and the shape recovery ratio (R_r) of CG@MF/PEG specimens could be calculated

$$R_f = \frac{\theta_1}{180} \times 100\% \quad (3)$$

$$R_r = \frac{\theta_1 - \theta_2}{\theta_1} \times 100\% \quad (4)$$

where θ_1 represents the temporary angle, and θ_2 represents the recovery angle.

An HSX-F300 xenon lamp (China) with a filter lens was applied to investigate the light-to-thermal energy transformation performance of CG@MF/PEG PCMs. The efficiency (η) of light-to-thermal energy storage was obtained according to eq 5

$$\eta = m \cdot \Delta H / (S \cdot P \cdot t) \quad (5)$$

where m , ΔH , and S represent the weight, melting enthalpy, and surface area of the specimens, respectively. P represents the irradiation power of the xenon lamp, and t represents the time taken for the light-driven phase transition.

The electric-to-thermal energy transformation measurement was performed using the DC power supply. The temperature variations of

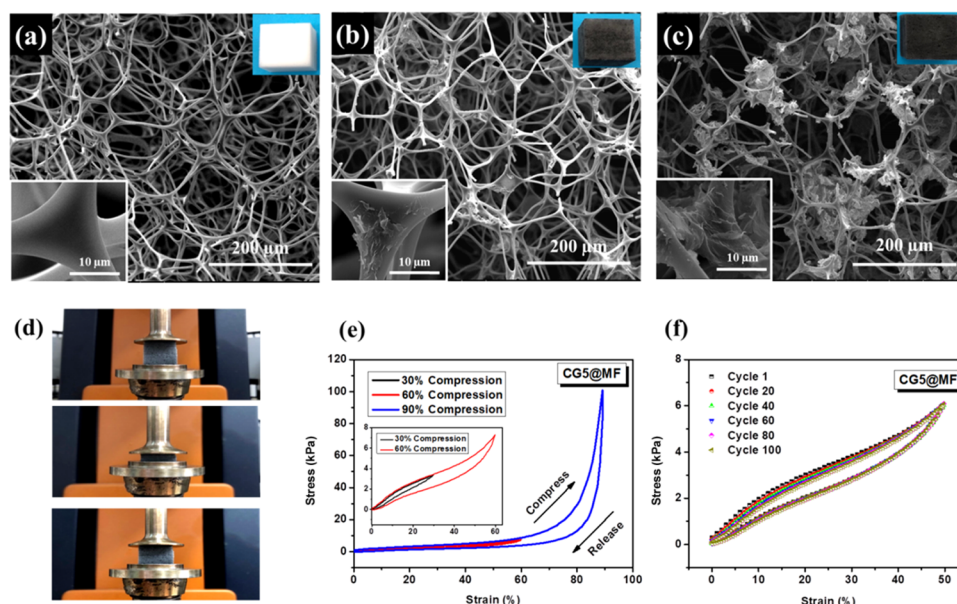


Figure 3. SEM characterization of (a) pure MF, (b) CG1@MF, and (c) CG5@MF sponges, respectively. (d) Digital photographs showing the compression–recovery behavior of the CG5@MF sponge (90% strain). (e) Compressive stress–strain curves of the CG5@MF sponge with various strains of 30, 60, and 90%. (f) Cyclic compressive stress–strain curves of the CG5@MF sponge during 100 cycles (50% strain).

samples were recorded via a thermograph and a data collecting system. The electrical conductivity of CG5@MF/PEG PCM was measured by the Keithley digital multimeter (DAQ6510). A digital camera was used to record the self-adhesion process of the fractured CG5@MF/PEG PCM stimulated by the heat source, light source, and electric source. The mechanical properties before and after the self-adhesion of CG5@MF/PEG PCM were tested by dynamic thermomechanical analysis (DMA, TA Q800). The samples were tested at 25 °C with a tensile speed of 3 N min^{−1}.

3. RESULTS AND DISCUSSION

3.1. Characterization of CNF/GNP Hybrid and CG@MF Sponges. In this work, CNF was used as a dispersant for GNP in water. The dispersion of CNF, GNP, and CNF/GNP in aqueous solution is first studied. As shown in Figure 2a, the CNF suspension is homogeneous and stable because of the existence of hydrophilic carboxyl groups. Pristine GNP sheets precipitate at the bottom after 24 h due to the lack of functional groups. However, the CNF/GNP hybrid suspension (with the weight ratio of CNF/GNP = 1:9) exhibits a stable dark state without any agglomeration, indicating the improved dispersion of GNP in water by adding CNF. To further confirm the effect of CNF-assisted GNP dispersion, the three suspensions were characterized by UV–vis absorbance spectroscopy, and the results are shown in Figure 2b. The peak at 269 nm is attributed to the $\pi \rightarrow \pi^*$ transition of aromatic C–C bonds.³⁸ It is clearly seen that the adsorption intensity at 269 nm of CNF/GNP hybrid solution is much larger than that of the pristine GNP solution at the same GNP content. This phenomenon is ascribed to the fact that abundant GNPs are suspended in the CNF/GNP hybrid solution, while many GNPs are precipitated in pure GNP solution. Therefore, it can be concluded that CNF plays a positive role in the homogeneous dispersion of GNPs in water. Because of the long-range electrostatic repulsion interactions among CNFs as well as the short-range hydrophobic interactions between the GNPs and the specific crystalline faces of CNFs, GNPs can be stably dispersed in aqueous solution with the aid of CNFs.⁴² The FTIR spectra further confirm the combination of CNFs

and GNPs. For the CNF/GNP hybrids, the characteristic FTIR peak at 1601 cm^{−1} belongs to the stretching vibration of –COO– groups in CNF, suggesting the successful loading of CNF on GNP sheets (Figure 2c). Direct morphologies for the difference between GNP and CNF/GNP suspensions were investigated by SEM. Pristine GNP sheets are aggregated in a large size (Figure 2d), while the CNF/GNP hybrids are well-dispersed without any obvious agglomeration (Figure 2e). The SEM image with high magnification shows that abundant CNFs are anchored on the surface of GNP sheets (Figure 2f). The morphology of CNF is displayed in Figure S1. CNF exhibits a one-dimensional (1D) fibrous feature with lengths of 3–5 μ m and diameters of 2–5 nm. The results mentioned above prove that CNF can serve as a good dispersant for GNP in the aqueous system.

The homogeneous dispersion of the CNF/GNP hybrid suspension enables the facile fabrication of graphene-based composites. The CG@MF sponges were prepared through coating the CNF/GNP hybrids onto the skeletons of MF sponges via a dip-coating method. Different from the neat MF sponge that is white in color, all of the CG@MF sponges exhibit uniform black appearance, suggesting that the CNF/GNP hybrids are successfully permeated into the MF sponges (Figure 3a–c). The content of CNF/GNP hybrids can be adjusted by changing the dip-coating cycle. It is obviously found that the color of CG@MF sponges becomes darker with adding the dip-coating cycle. The morphological structures of neat MF and CG@MF sponges with different immersion times were investigated by SEM. All of the sponges show a porous and 3D interconnected framework. The neat MF has a smooth surface and exhibits a porous structure, which has a big pore size (100–150 μ m) (Figure 3a). After immersion in the CNF/GNP hybrid suspension for one time, the CNF/GNP hybrids are tightly coated and randomly connected along the skeletons of MF sponges, leading to the formation of a thin GNP layer (Figure 3b). When the immersion time increases to 5 times, the CNF/GNP hybrids stack one by one along the MF skeleton to construct a continuous network, resulting in a

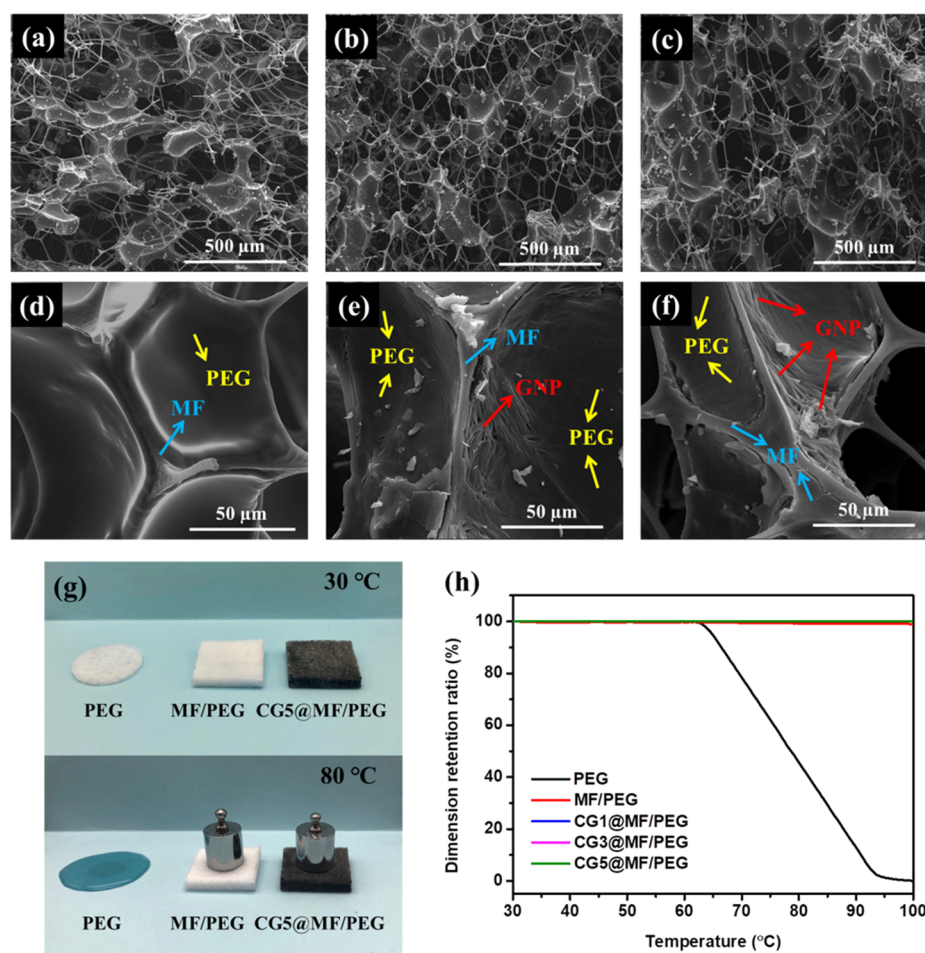


Figure 4. SEM images of MF/PEG (a, d), CG1@MF/PEG (b, e), and CG5@MF/PEG (c, f) at different magnifications. (g) Visual photos illustrating the difference in the encapsulation effect of the PEG, MF/PEG, and CG5@MF/PEG at 30 and 80 °C. (h) Dimension retention ratio curves of PEG, MF/PEG, and CG@MF/PEG during the heating process.

rougher surface with a wrinkled feature (Figure 3c). Moreover, the CG@MF sponges possess outstanding compressibility and recoverability because of the high flexibility of MF sponges. The CG5@MF sponge can be compressed to a strain of 90% and rapidly recovers to its original shape after releasing the compression load (Figure 3d). Figure 3e displays the compressive stress–strain curves for the CG5@MF sponge under different compression strains (30, 60, and 90%), suggesting that the sponge has excellent flexibility. Besides, this property remains nearly constant after 100 compression cycles, confirming its excellent mechanical stability (Figure 3f). This performance permits the CG@MF sponge to be applied as an ideal supporting scaffold for the preparation of flexible FSPCMs.

3.2. Morphology and Encapsulation Properties of CG@MF/PEG PCMs. The CG@MF sponges have open macropores, which are beneficial for the impregnation of PEG melts. Figure 4 shows the SEM images of the MF/PEG, CG1@MF/PEG, and CG5@MF/PEG PCMs. The three representative PCMs exhibit no remarkable difference in their overall morphology (Figure 4a–c). The absorbed PEG mainly fills in the macropores or adheres along the skeletons of the sponges. For the CG@MF/PEG PCMs, the pores become more impact than those of MF/PEG PCM, since both CNF/ GNP and PEG occupy the pores (Figure 4d–f). These SEM images demonstrate that the MF and CG@MF sponges

provide porous scaffolds to effectively absorb the PEG macromolecules. FTIR and X-ray diffraction (XRD) measurements were conducted to investigate the structure of the CG@MF/PEG PCMs (Figure S2). The characteristic peaks of PEG have no obvious change with the addition of CG@MF sponges, suggesting that physical interactions mainly exist between CG@MF and PEG. Herein, it should be noted that the inner pores of CG@MF are not fully and homogeneously filled by PEG due to the physical interactions between CG@MF and PEG. It is necessary to modify the MF sponge to enhance the chemical interactions between MF and PEG, and thus makes PEG homogeneously filled in the macropores of MF sponges.

Furthermore, these sponges with 3D interconnected networks are ideal supporting materials to prohibit the leakage of the melted PEG. Figure 4g shows the difference in shape stability of the PEG, MF/PEG, and CG5@MF/PEG PCMs at 30 and 80 °C. All of the three representative PCMs keep their shape stable at 30 °C, which is below the melting temperature (T_m) of PEG. However, when the samples are heated to 80 °C, they show totally different shape stability. Neat PEG melts and liquidizes almost completely at 80 °C, but MF/PEG and CG5@MF/PEG still remain in their original shape without any PEG leakage. Even under compression conditions of 20 g of load, MF/PEG and CG5@MF/PEG retain their shape well, indicating that MF plays an important role in supporting PEG

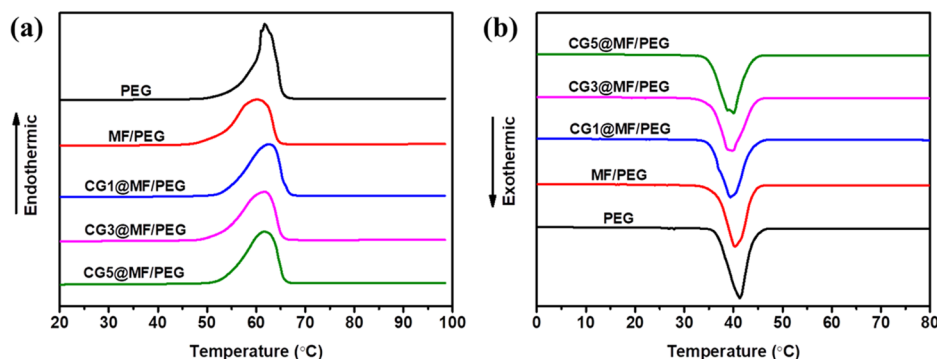


Figure 5. DSC curves of PEG, MF/PEG, and CG@MF/PEG. (a) DSC heating scans and (b) DSC cooling scans.

melts. The dimension retention ratios of PEG, MF/PEG, and CG@MF/PEG were quantitatively measured using the dynamic rheometer. As shown in Figure 4h, the dimension of neat PEG starts to decrease sharply when the temperature reaches 60 °C. In contrast, the dimensions of MF/PEG and CG@MF/PEG remain constant from 30 to 100 °C, resulting from the effective encapsulation of MF sponge. This phenomenon demonstrates that the MF and CG@MF sponges with 3D interconnected frameworks are suitable supporting scaffolds to avoid the leakage of melted PEG.

3.3. Phase Change Performances of CG@MF/PEG PCMs. The phase transition temperatures and latent heat are important for PCMs to determine their thermal energy storage properties. Figure 5 shows the melting and freezing behaviors of PEG, MF/PEG, and CG@MF/PEG PCMs. Neat PEG shows an endothermic peak at 61.8 °C and an exothermic peak at 41.2 °C, which are corresponded to the melting phase transition and the crystallization phase transition, respectively. The T_m s and T_c s of MF/PEG and CG@MF/PEG PCMs are very similar to those of neat PEG. This is ascribed to the fact that there is no chemical reaction between CG@MF sponges and PEG during the vacuum impregnation process. The detailed thermal parameters such as T_m , melting enthalpy (ΔH_m), crystallization temperature (T_c), and crystallization enthalpy (ΔH_c) are listed in Table 1. The ΔH_m of neat PEG is

Table 1. Phase Change Characteristics of PEG, MF/PEG, and CG@MF/PEG

sample	T_m (°C)	T_c (°C)	ΔH_m (J g ⁻¹)	ΔH_c (J g ⁻¹)	latent heat retention (%)
PEG	61.8	41.2	191.2	200.5	
MF/PEG	60.1	40.3	178.3	190.7	95.1
CG1@MF/PEG	62.4	39.6	182.4	192.8	96.2
CG3@MF/PEG	61.9	39.4	176.6	191.7	95.6
CG5@MF/PEG	61.7	39.5	178.9	192.4	96.0

191.2 J g⁻¹, and a small decline in the ΔH_m is found for MF/PEG (178.3 J g⁻¹) and CG5@MF/PEG (178.9 J g⁻¹). The mass fraction of PEG in the CG5@MF/PEG is calculated as 95.2%. Theoretically, the ΔH_m of CG5@MF/PEG is about 182.0 J g⁻¹ by multiplying the ΔH_m of PEG with its mass fraction because the CG5@MF sponge does not participate in the phase change, which is nearly equal to the actual ΔH_m of CG5@MF/PEG (178.9 J g⁻¹). This result indicates that the ΔH_m of PEG has a negligible loss after the CG5@MF sponge is incorporated. Therefore, these CG@MF/PEG PCMs with high latent heat can be utilized as efficient PCMs for heat storage.

3.4. Thermal Conductivity of CG@MF/PEG PCMs. For the practical application of PCMs, the thermal conductivity is a primary factor, which shortens the time needed for energy storage.^{43,44} It is well-known that GNP is an excellent conductive nanofiller. The thermal conductivity coefficient of single-layered graphene can reach as high as 5000 W m⁻¹ K⁻¹.⁴⁵ To investigate the thermal conductivity of CG@MF/PEG PCMs, the surface temperature variations of MF/PEG and CG@MF/PEG with different heating times were recorded by an infrared thermal imager. The samples were placed on a hotplate at 80 °C. As shown in Figure 6a, the color of CG@MF/PEG PCMs changes quicker than that of MF/PEG PCM, suggesting that the heating rate is obviously accelerated and GNP sheets can remarkably enhance the heat transfer rate of PCMs. More importantly, the degree of temperature increment is extremely affected by the content of GNP in the CG@MF/PEG PCMs. Figure 6b gives the temperature–time plots of MF/PEG and CG@MF/PEG during the heating process. It is clearly seen that the surface temperature of CG@MF/PEG PCMs is higher than that of MF/PEG PCM at the same heating time. The higher the GNP content in the PCMs, the more pronounced the temperature increase. The thermal conductivity of CG5@MF/PEG is 0.26 W m⁻¹ K⁻¹, which is 188.9% higher than that of MF/PEG (0.09 W m⁻¹ K⁻¹). Due to the improved thermal conductivity of CG@MF/PEG PCMs, the heat diffusion in the PCMs is accelerated, and thus the temperature variation is significantly enhanced. It should be emphasized that the weight ratio of PEG to CG@MF is 20:1. The macropores of CG@MF sponges are not fully filled with PEG. This is quite different from other FSPCMs, where the inner pores are always entirely occupied by PCMs. It is known that air is a poor conductor of heat. Thus, the thermal conductivity of CG@MF/PEG is relatively low due to the existence of internal voids. Nevertheless, the GNP network formed along the MF skeleton provides a continuous thermal conductive path to enhance the thermal conductivity of PCMs, which is significant for the effective thermal management.

3.5. Thermal-Driven Shape Memory Properties of CG@MF/PEG PCMs. Apart from the phase change enthalpy and thermal conductivity, the strong rigidity of FSPCMs will lead to the easy brittle failure, which is bad for the practical installation of FSPCMs.⁶ Thus, FSPCMs with flexibility are highly needed to solve the installation difficulty. In this work, the elastic CG@MF sponges are used as supporting scaffolds to fabricate the flexible FSPCMs. The combination of PEG with CG@MF allows the deformation and recovery of shape above the T_m of PEG as well as the shape fixation below the T_c of PEG. As shown in Figure 7a, the thermal-driven shape

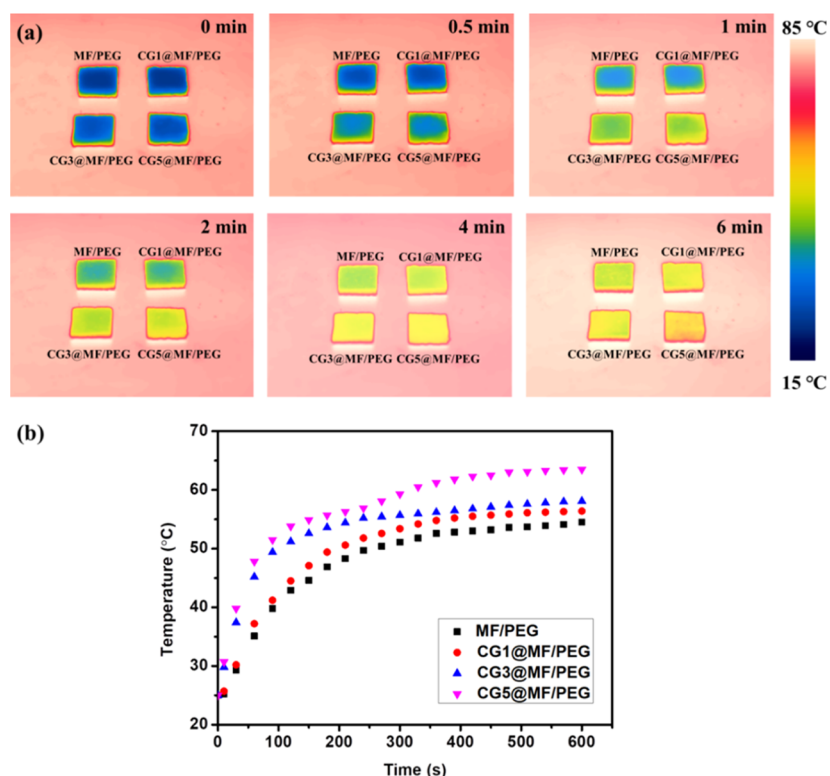


Figure 6. (a) Infrared thermography photos showing the surface temperature increments of MF/PEG and CG@MF/PEG PCMs at different heating times. (b) Temperature changes of MF/PEG and CG@MF/PEG PCMs with increasing heating time.

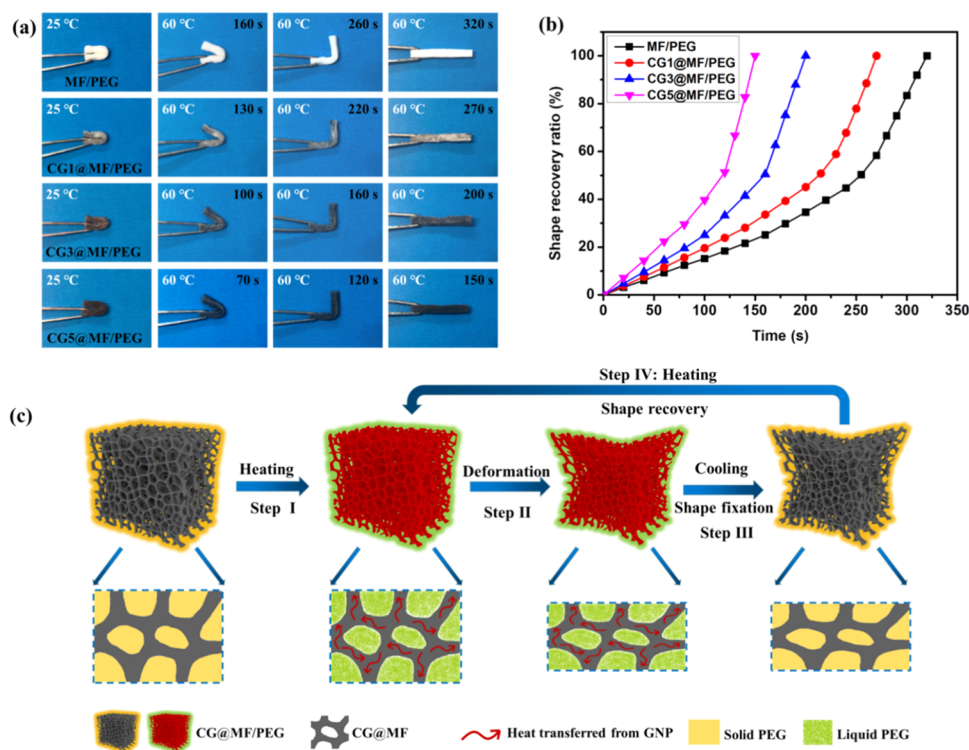


Figure 7. (a) Digital photographs illustrating the thermal-driven shape memory effects of MF/PEG and CG@MF/PEG PCMs. (b) Variations of the shape recovery ratio with increasing heating time. (c) Thermal-driven shape memory mechanism of CG@MF/PEG PCMs.

memory performances of MF/PEG and CG@MF/PEG PCMs are visually demonstrated. The straight samples were easily bent into a “U” shape at 60 °C and then quickly placed it in liquid nitrogen. The deformed shape was well fixed at 25 °C

and no obvious shape recovery was seen when the samples were placed in air for 2 h, indicating that the MF/PEG and CG@MF/PEG PCMs display good shape fixing performances. Since the deformation is not in a steady state above the T_m of

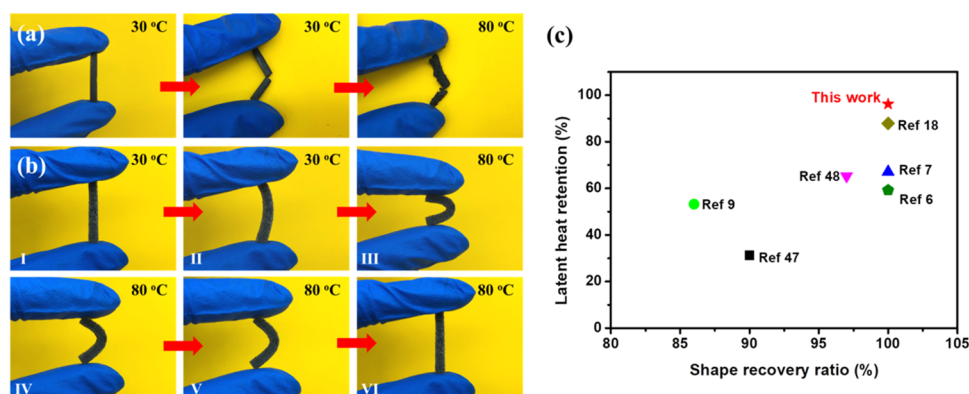


Figure 8. (a) Illustration of bending deformation for PEG-based PCMs supported by graphene aerogels. (b) Bending deformation for CG@MF/PEG PCMs at 30 °C and shape recovery at 80 °C. (c) Comparisons of the shape recovery ratio and latent heat retention of CG@MF/PEG PCMs with other flexible FSPCMs reported in the literature.

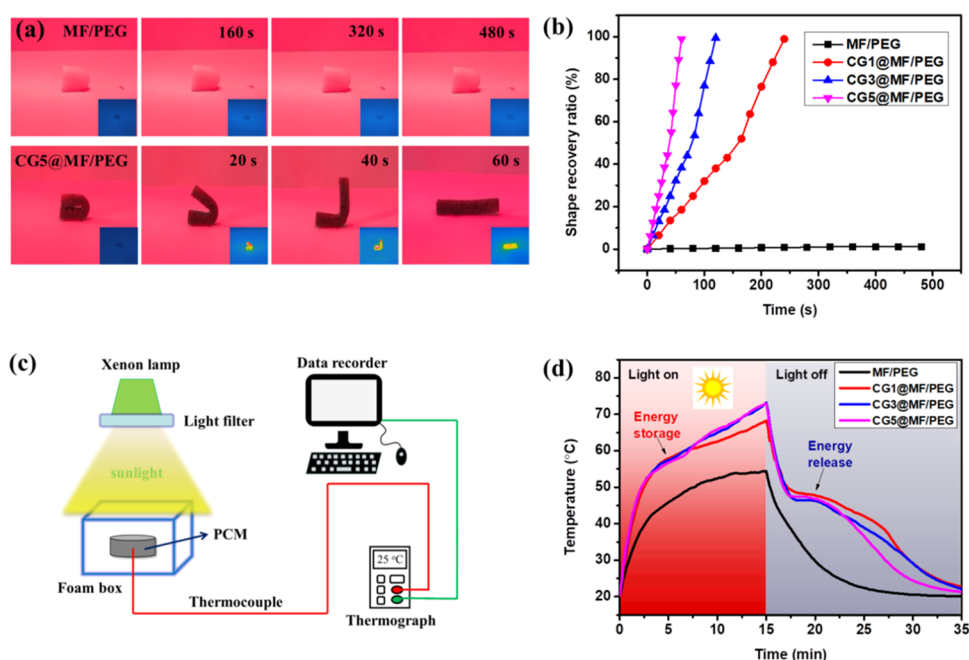


Figure 9. (a) Digital photographs showing the NIR light-driven shape recovery effects of MF/PEG and CG5@MF/PEG. The lower right images are the infrared thermal images showing the real-time temperature. (b) Quantitative variations of the shape recovery ratio with increasing irradiation time. (c) Experimental setup of the light-to-thermal energy transformation measurement device. (d) Temperature variation plots with and without the simulated solar illumination (xenon lamp).

PEG, it is necessary to fix the deformed shape at 25 °C (below the T_c of PEG). Actually, the shape can be well fixed simply by cooling below the T_c of PEG. It would just take longer (for about 300 s) to freeze in natural cooling. When the samples with fixed “U” shape were placed in the hotplate (80 °C), they could all quickly recover to their original straight shape. More importantly, the time of shape recovery decreases with the addition of CNF/GNP hybrids (Figure 7b). The shape recovery time of CG5@MF/PEG is 150 s, which is much lower than that of MF/PEG (320 s) and CG1@MF/PEG (270 s). This is ascribed to the fact that the increased CNF/GNP content accelerates the thermal transfer in PCMs, and thus the time for reaching the T_m of PEG is shortened. According to the above results, the prepared CG@MF/PEG PCMs exhibit coinstantaneous improved thermal energy storage and thermal-driven shape memory properties, which are beneficial for broadening the applications of PCMs in a confined space.

The thermal-driven shape memory mechanism of CG@MF/PEG PCMs is further illustrated in Figure 7c. The elasticity of CG@MF sponges provides the shape recovery force, while fixing and unfixing the temporary shape are regulated by the crystallization and melting of PEG chains. In detail, the CG@MF/PEG PCMs transform into a flexible block upon heating above the T_m of PEG (step I), and they can be deformed into another shape under the external force (step II). Then, the CG@MF/PEG PCMs are quickly cooled to 25 °C (below the T_c of PEG), and PEG crystallizes in the inner pores of CG@MF sponges. Due to the formation of rigid PEG crystals, the modulus of CG@MF/PEG PCMs at 25 °C is much higher than the elastic force of CG@MF sponges. After releasing the external force, the temporary shape is capable of being fixed at 25 °C (step III). Finally, when the CG@MF/PEG PCMs are reheated above the T_m of PEG, PEG crystals melt again. In this case, the stored elastic force of CG@MF sponges is rapidly

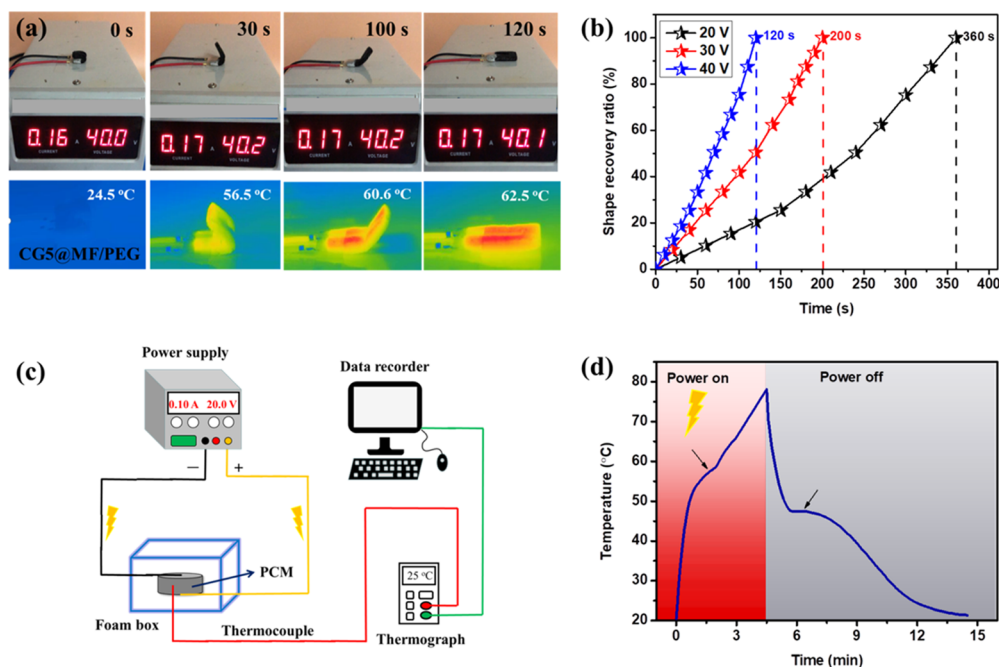


Figure 10. (a) Digital photographs and infrared thermography photos displaying the electro-driven shape recovery behavior of CG5@MF/PEG under 40 V. (b) Shape recovery ratio variations of the representative CG5@MF/PEG at different voltages. (c) Experimental setup of the electro-thermal energy conversion measurement system. (d) Temperature variation curve of the representative CG5@MF/PEG with or without the power supply.

released and provides a driving force for the CG@MF/PEG PCMs to achieve the recovery of initial shape (step IV).

Many literature studies have reported the excellent mechanical properties of FSPCMs based on graphene aerogel. For instance, Yu et al. and Wang et al. reported that their FSPCMs could maintain their shapes well above the T_m of PCM, even compressed by a weight of 500 g.^{26,46} However, the strong rigidity of FSPCMs may lead to the easy brittle failure under huge bending, which inevitably leads to the installation difficulty and poor surface contact with devices. In this work, our aim is to prepare a flexible FSPCM with excellent latent heat retention and shape memory properties. The differences in the thermal-induced flexibility properties of conventional FSPCMs and CG@MF/PEG PCMs are shown in Figure 8a,b. The experiments were conducted at 30 and 80 °C, which was under and above the T_m of PEG, respectively. For the conventional FSPCMs supported by graphene aerogel (Figure 8a), the sample shows strong rigidity and presents brittle failure under the external force. In contrast, the FSPCMs supported by CG@MF sponges exhibit superior flexibility based on the observation of a large degree of bending deformation (Figure 8b). This phenomenon is ascribed to the different softness of supporting materials. Unlike the brittle graphene aerogel, the elastic CG@MF sponge enables invertible bending deformation and recovery of CG@MF/PEG PCMs. Besides, the low-density CG@MF sponges have a strong adsorption capacity for PEG. Its absorption ratio is as high as 2000 wt %, thus endowing the CG@MF/PEG PCMs with high latent heat and superior heat energy storage capability. To the best of our knowledge, the flexible FSPCMs reported in this work exhibit both superior latent heat retention as well as a high shape recovery ratio, compared with other flexible FSPCMs in the literature (Figure 8c).^{47,48}

3.6. Light-Driven Shape Memory and Light-to-Thermal Energy Conversion Properties. The CG@MF

sponge combines the advantages of both MF and GNP, which endows the CG@MF sponge not only with excellent photoabsorption ability but also with electrical conductivity, thus showing great potential for applications in light/electro-to-thermal energy storage and light/electro-driven shape memory fields. The light-driven shape memory effect of CG@MF/PEG PCMs was investigated under a NIR lamp with a constant intensity of 20 mW cm⁻². From Figures 9a and S3, no remarkable shape recovery is observed for MF/PEG, because MF and PEG can hardly capture photon. In contrast, the CG5@MF/PEG has the ability to recover from the temporary “U” shape to the initial straight shape (in 60 s). The quantitative variations of the shape recovery ratio with increasing irradiation time are shown in Figure 9b. Under NIR light radiation, the shape recovery speed increases in proportion to the content of CNF/GNP hybrids. This behavior results from the effective photon capture and nanoheating effect of GNP sheets,²⁸ which convert NIR light energy into heat and allow the CG@MF/PEG PCMs to increase their temperature rapidly. When the temperature reaches the T_m of PEG, the resilience force of CG@MF sponges induces the instant shape recovery without the prevention of PEG crystals.

The measurements of light-to-thermal energy transformation were conducted under a simulated solar illumination (xenon lamp) with an intensity of 100 mW cm⁻². A thermograph and a data collecting device were used to record the temperature variations inside the PCMs, as displayed in Figure 9c. The temperature variation curves with or without the simulated solar illumination are shown in Figure 9d. Compared with MF/PEG, the internal temperature of CG@MF/PEG PCMs increases more quickly. As the irradiation time increases, the temperature plateau of CG@MF/PEG PCMs emerges between about 54.2 and 58.5 °C, indicating that the CG@MF/PEG PCMs experience a solid–liquid phase transition

period and the solar energy is transferred into heat energy as latent heat.⁴⁹ Once the xenon lamp is shut down, the temperature of CG@MF/PEG PCMs exhibits a sharp decline in the initial stage. Then, a freezing plateau between 47.5 and 43.9 °C emerges, indicating that the stored thermal energy is released via the crystallization of PEG.^{50,51} In addition, due to the enhanced thermal conductivity, less time is needed for the CG5@MF/PEG to achieve 20 °C. Nevertheless, the MF/PEG displays no obvious melting and freezing plateaus. The light-to-thermal energy storing efficiency of the CG5@MF/PEG is about 89.88%. These results prove that GNP sheets are effective photon absorbers and nanoheating units, which can convert solar energy into thermal energy and then transfer thermal energy across the CG@MF/PEG PCMs.

3.7. Electro-Driven Shape Memory and Electro-to-Thermal Energy Conversion Properties. The CG@MF/PEG PCMs are not only light-responsive but also exhibit electro-driven shape memory performance and electrical energy storing ability. It is known that PEG is an insulated polymer with an electrical conductivity of about 10^{-14} S m⁻¹.²⁶ With the incorporation of CNF/GNP hybrids, the electrical conductivity of the CG5@MF/PEG is 6.19 S m⁻¹, an increase of 15 orders of magnitude compared to that of neat PEG. Due to its moderate electrical resistivity, CG5@MF/PEG is capable of being activated under a low electric voltage to exhibit the shape memory effect. Figure 10a shows the electro-driven shape recovery behavior of CG5@MF/PEG under 40 V. The temperature rises quickly with the increasing conduction time and reaches the T_m of PEG in about 100 s. The CG5@MF/PEG shows a faster shape recovery process, and the temporary shape is fully recovered in a short time (120 s). The shape recovery ratio variations of the CG5@MF/PEG at different voltages are shown in Figure 10b. As expected, increasing the voltage brings about a more apparent enhancement of the shape recovery speed in a shorter time. This phenomenon is ascribed to that of a higher voltage results in more Joule heat in unit time. When the generated heat causes the temperature above the T_m of PEG, the resilience of CG@MF sponge induces the shape recovery.

Furthermore, the electro-to-thermal conversion and storage effect can be achieved based on the 3D continuous conductive network of GNP. The experimental setup of the electro-to-thermal energy conversion measurement system is illustrated in Figure 10c, and the temperature–time curve of the reprehensive CG5@MF/PEG is shown in Figure 10d. Similar to the solar-to-thermal energy conversion, a melting plateau (53.5–58.3 °C) emerges when a voltage (20 V) is applied, suggesting the electro-to-thermal energy transformation and storage process. When the power is cut off, the temperature decreases acutely until a cooling plateau (43.5–47.6 °C) emerges, which indicates that the heat saved during the power-on period is released. The CG5@MF/PEG possesses an electro-to-thermal energy conversion efficiency of about 66.13%.

3.8. Multiresponsive Self-adhesion Properties. It should be noted that the flexible FSPCMs may fracture after repeated deformation or destruction, resulting in a sharp decline in their sustainability and lifetime. Therefore, there is an urgent need to explore the self-adhesion properties that can be repaired after mechanical destruction. To evaluate the self-adhesion effect, the CG5@MF/PEG sample was cut into two halves by a scalpel. Then, the two halves were placed together without pushing. The fractured sample was repaired by direct

heating (Figure 11a), NIR light (Figure 11b), and electricity (Figure 11c), respectively. Interestingly, all of the two halves

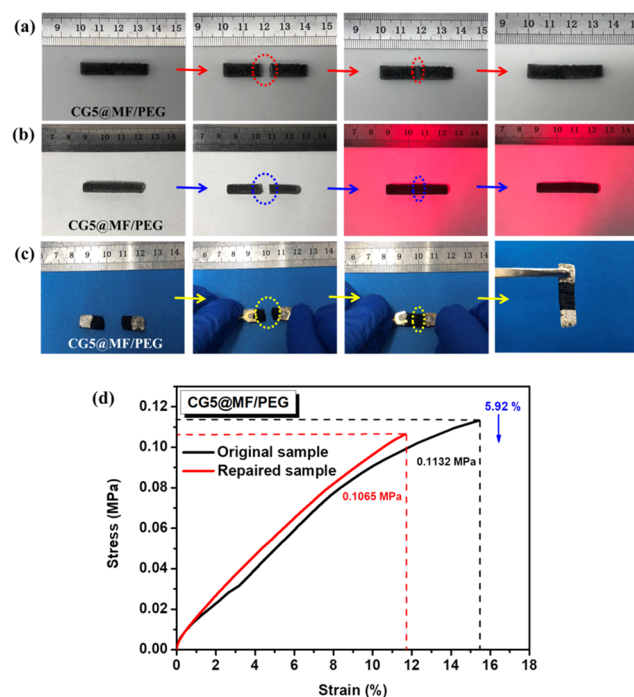


Figure 11. Digital photographs showing the self-adhesion behaviors of the CG5@MF/PEG under (a) direct heating, (b) NIR light, and (c) electricity. (d) Tensile stress–strain curves of the original and repaired CG5@MF/PEG under NIR light irradiation for 60 s.

revealed to merge into one entire piece after repairing for 60 s. The repairing efficiency was evaluated by testing the tensile strength of the original and repaired samples. Taking the sample repaired by NIR light as an example (Figures 11d and S4), the stress at failure (0.107 MPa) of the repaired sample almost completely achieves its original stress (0.113 MPa) after 60 s. The above results demonstrate that the CG@MF/PEG PCMs can realize a fast self-adhesion effect using different stimuli (heat, light, and electricity). Herein, GNP sheets play an important role in converting the light or electricity into thermal energy and transferring the thermal energy into the PCMs. Then, the internal temperature of PCMs increases quickly, and the self-adhesion effect is achieved through the wetting, diffusion, and re-entanglement of PEG chains over the T_m of PEG. After removing the stimuli, PEG crystallizes at room temperature (25 °C) and these PEG crystals bridge the fracture area to merge the two pieces into one.^{52,53} However, the sample will break when the temperature is above the T_m of PEG since MF is permanently broken. Thus, the repaired sample can only be used at room temperature for some time. However, the CG@MF/PEG PCMs with a self-adhesion ability may have a prolonged service life in practical utilization.

3.9. Potential Application of a Shape-Adaptable and Thermal-Insulating Container. Considering the excellent shape memory performance as well as light/electro-to-thermal energy conversion performance, the CG@MF/PEG PCMs may show advantages in some specific fields. For example, the PCM can be used as a shape-adaptable and thermal-insulating container, which captures light energy and saves the converted heat energy. As shown in Figure 12a, the cubic container can be compressed into a smaller shape and then recovered to its

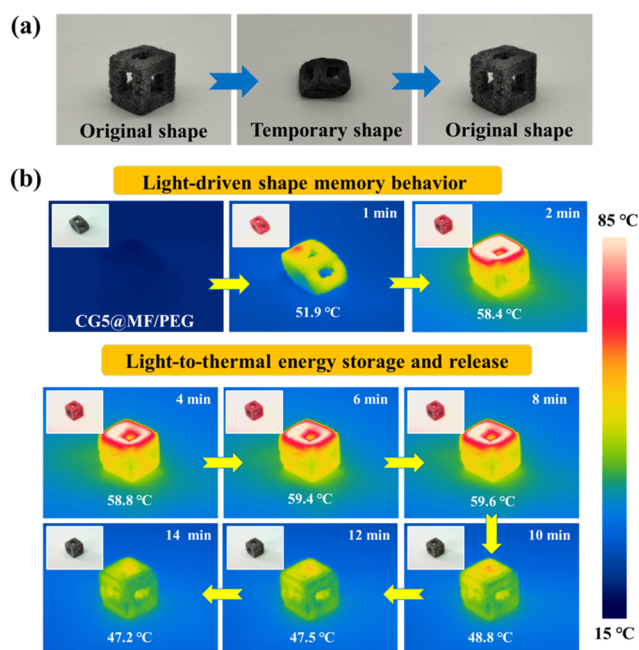


Figure 12. (a) Digital photographs showing the shape memory behaviors of a container made of CG5@MF/PEG. (b) Infrared thermal images displaying the light-driven shape recovery performance and the subsequent light-to-thermal energy storage and release.

original shape under light irradiation, suggesting its good shape adaptability. The infrared thermal imager was applied to investigate the light-driven shape recovery performance and the succeeding light-to-thermal energy conversion performance of the container. As shown in Figure 12b, the container displays a remarkable temperature increase under the irradiation of an NIR lamp. The temperature quickly reaches 58.4 °C, and the compressed shape recovers its original shape within 2 min. In the following irradiation process, the temperature of the container remains constant at about 59 °C for 6 min, resulting from the high energy storage density of the container. In this case, the container made of CG5@MF/PEG absorbs light energy, converts it into thermal energy, and saves thermal energy as latent heat. Moreover, when light irradiation is removed, the container still maintains a moderate temperature (about 48 °C) for 6 min, further confirming the excellent light-to-thermal energy storing ability of CG5@MF/PEG. Therefore, the container can keep its temperature constant for a long time through the storage and release of light-to-thermal energy. In the devices with confined spaces, the shape-adaptable and thermal-insulating container can be installed more easily, deployed into the desired shape under light or electricity, and keeps its internal temperature constant. For example, the CG@MF/PEG PCMs can be applied for the thermal management of plastic tubes under overheating and overcooling circumstances (Figure S5). The flexibility of CG@MF/PEG PCMs allows a good surface contact between PCMs and tubes. Under overheating conditions, the CG@MF/PEG PCMs with high latent heat can work as flexible heat absorbers to store the excess thermal energy in the form of latent heat. Under extremely cold conditions, the CG@MF/PEG PCMs with light-responsibility can convert solar energy into heat energy and generate urgent heat to tubes.

4. CONCLUSIONS

In summary, the multiresponsive shape-adaptable FSPCMs with prominent light/electro-to-thermal energy transformation and storage capability were fabricated by infiltrating CG@MF sponges with PEG. The CG@MF sponges were fabricated by the assisted dispersion of CNF and the repeated dip-coating with CNF/GNP hybrids. The MF supported GNP network imparts the CG@MF/PEG PCMs with enhanced thermal conductivity, light/electro-to-thermal energy storing ability, and light/electro-driven shape variation capability. In addition, the CG@MF/PEG PCMs can exhibit a good self-adhesion effect under three different stimuli. One potential application of a light-sensitive, shape-adaptable, and thermal-insulating container is further explored. The multifunctional PCMs show promising potential applications in many energy storage devices.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.9b16612>.

Digital photograph of the CNF suspension; TEM image and AFM image of CNF; FTIR spectra of CG5@MF, PEG, MF/PEG, and CG@MF/PEG; XRD patterns of PEG, MF/PEG, and CG@MF/PEG PCMs; digital photographs showing the NIR light-driven shape recovery performances of CG@MF/PEG PCMs; tensile stress–strain curves of the original and repaired CG5@MF/PEG PCM under the NIR light irradiation for three specimens; infrared thermal images showing the application in the thermal management of plastic tubes under overheating and overcooling circumstances (PDF)

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Notes

The authors declare no competing financial interest.

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